Netflix Data Analysis Project

A Comprehensive Analysis Using Python, Google Cloud, and Tableau

1. Objective

The primary goal of this project is to analyze Netflix's dataset to uncover insights about its content library, including trends in content creation, most popular genres, and content distribution across different regions and time periods. This project also aims to build a visually engaging dashboard for storytelling using Tableau.

2. Problem Statement

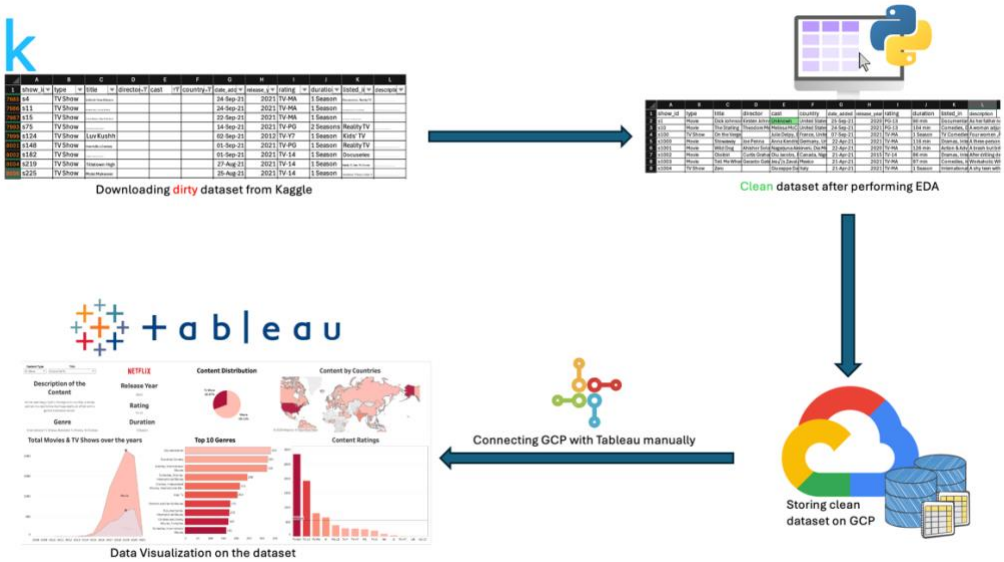
- Netflix's extensive library lacks an easily accessible summary of trends for informed decision-making.
- Challenges included integrating cloud storage with Tableau and ensuring clean and consistent data for analysis.
- Deciding on the most suitable cloud platform (e.g., GCP, IBM Cloud, Oracle Cloud Infrastructure) required detailed comparisons and discussions.

3. Data Architecture Diagram

The diagram should illustrate the flow of data from its source (raw CSV file) to Google Cloud Storage, followed by data cleaning and analysis in Python, and finally its visualization in Tableau.

Components:

a. Raw Dataset from Kaggle (netflix_dataset.csv)
b. Python for Data Cleaning and Exploratory Data Analysis (EDA)
c. Google Cloud Platform (Storage for cleaned data)
d. Tableau for Dashboard Design



4. Roles and Responsibilities Matrix

Responsibility	Karan	Shrajna	Aman
Brainstorming & Planning	Discussed project scope and tools. Provided ideas on data structure and tools.	Collaborated in brainstorming. Suggested key data visualization ideas. Identified visual elements for Tableau.	Contributed to brainstorming. Highlighted data cleaning needs. Proposed key EDA methods.
Tool Research	Explored cloud options. Conducted comparisons between GCP, IBM Cloud, and Oracle Cloud.	Researched Tableau's functionality and features.	Researched cleaning techniques and visualization standards. Suggested Python libraries for EDA.
Data Cleaning	Shared raw data with the team. Participated in discussions on cleaning methods.	Collaborated on defining cleaned data requirements. Ensured cleaning aligned with visual goals.	Led cleaning process. Worked on structuring the cleaned data.
EDA	Contributed resources for analysis.	Provided suggestions for EDA insights relevant to visualization.	Led EDA efforts. Generated insights from the cleaned data.
GCP Management	Finalized GCP. Uploaded cleaned data to GCP Storage. Connected GCP with Tableau, manually	Supported with data upload.	Provided feedback during GCP testing. Ensured data compatibility post upload.
Tableau Dashboard	Assisted in deciding visual elements and filter designs.	Designed and built the Tableau dashboard. Finalized charts and interactions. Used student ID for Tableau access.	Reviewed dashboards and provided insights for refinement. Tested dashboard usability.

Summary

• The project required extensive collaboration across Python, GCP, and Tableau tools.
• All three team members contributed significantly to all tasks, ensuring a well-rounded project.
• Accomplishments:

a. Karan handled cloud management and tool integration.
b. Shrajna led Tableau dashboard creation using a student ID.
c. Aman spearheaded EDA, ensuring the team had meaningful insights for visualization.

5. Steps of the Project

Step 1: Data Collection

• Source: Netflix dataset in CSV format from Kaggle.
• Reviewed dataset for relevance and completeness.

Step 2: Data Cleaning and EDA

• Cleaning involved handling missing values, ensuring consistent formats, and removing duplicates.
• EDA: Performed using Python libraries like Pandas and Matplotlib to uncover trends in Netflix's content library.

Exploratory Data Analysis (EDA) on Netflix Dataset

Below are the structured steps for EDA, written for inclusion in a report. Each step includes a brief explanation and key code snippets to provide clarity.

1. Import Libraries and Load the Dataset

To begin the analysis, we imported the necessary libraries and loaded the Netflix dataset into a Pandas DataFrame. This step ensures we can manipulate and analyze the data effectively.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the Netflix dataset
netflix_df = pd.read_csv("netflix_titles.csv")

# Display the first few rows
netflix_df.head()
```

2. Dataset Overview

We examined the structure and basic statistics of the dataset to understand its contents and dimensions.

```
# Basic information about the dataset
netflix_df.info()

# Summary statistics
print(netflix_df.describe())

# Shape of the dataset
print("Shape of the DataFrame:", netflix_df.shape)
```

3. Handle Missing Values
To ensure a clean dataset, missing values in critical columns like director, cast, and country were replaced with "Unknown." Rows with missing values in other columns were removed.

```
# Replace missing values with 'Unknown'
netflix_df['director'] = netflix_df['director'].fillna('Unknown')
netflix_df['cast'] = netflix_df['cast'].fillna('Unknown')
netflix_df['country'] = netflix_df['country'].fillna('Unknown')

# Drop rows with remaining missing values and reset the index
netflix_df.dropna(inplace=True)
netflix_df.reset_index(drop=True, inplace=True)
```

4. Rename Columns
To improve clarity, the type column was renamed to content type.

```
netflix_df.rename(columns={'type': 'content type'}, inplace=True)
```

5. Data Filtering and Subset Creation
We created specific subsets of the dataset for deeper analysis, such as identifying recently added TV shows and long-duration content.

- **Recently Added TV Shows**:

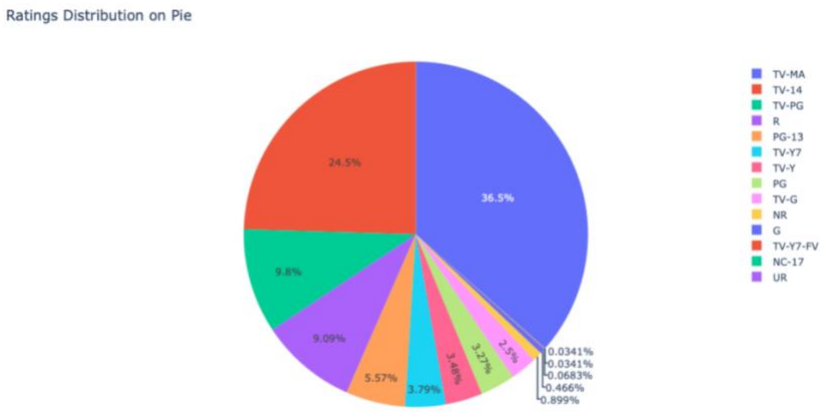
```
recent_tv_shows = netflix_df[
    (netflix_df['content type'] == 'TV Show') &
    (netflix_df['date_added'] > pd.Timestamp('2016-12-23'))
][['title', 'release_year', 'content type']]
```

- **Long-Duration Content**:

```
long_duration_content = netflix_df[
    netflix_df['duration'].str.contains(r'Season|[9-9]\d    min',
regex=True, na=False)
][['title', 'duration', 'content type']]
```

6. Visualization

Interactive visualizations were created using Plotly to provide insights into content distribution and ratings.



Key Insights

- The dataset provided critical information on Netflix content trends, such as content type, release year, and duration.
- Missing data handling and column renaming improved the dataset's usability for further analysis.
- Interactive visualizations highlighted the proportion of Movies vs. TV Shows and the distribution of content ratings.

Step 3: Cloud Integration

• Google Cloud Storage was chosen after evaluating alternatives.

Using Google Cloud Platform (GCP) for Data Storage and Manual Connection with Tableau Web

As part of this project, we leveraged Google Cloud Platform (GCP) to store and manage the dataset securely. To enable visualization in Tableau Web, we manually downloaded the processed dataset from GCP and uploaded it to Tableau. Below are the key steps we followed:

1. Storing and Managing Data in GCP

To store the cleaned dataset in GCP, we first configured the **Google Cloud SDK** to interact with GCP storage.

- **Set up Google Cloud SDK**:

We downloaded and installed the Google Cloud SDK by following the instructions provided on the Google Cloud CLIpage. After installation, we initialized the SDK with the following command:

```
gcloud init
```

This allowed us to authenticate our account, set the default project, and configure the CLI for seamless interaction.

- **Verify Access to Storage Buckets**:

To ensure access to the storage bucket, we listed all available buckets:

```
cloud storage buckets list
```

The bucket containing the cleaned Netflix dataset, *netflix-eda-bucket*, was confirmed.

```
(base) karanparmar@Karans-MacBook-Air ~ % gcloud storage buckets list

---
creation_time: 2024-11-18T03:40:01+0000
default_storage_class: STANDARD
generation: 1731901201288679427
location: US
location_type: multi-region
metageneration: 2
name: netflix-eda-bucket
public_access_prevention: enforced
rpo: DEFAULT
soft_delete_policy:
  effectiveTime: '2024-11-18T03:40:01.640000+00:00'
  retentionDurationSeconds: '604800'
storage_url: gs://netflix-eda-bucket/
uniform_bucket_level_access: true
update_time: 2024-11-28T18:16:37+0000
(base) karanparmar@Karans-MacBook-Air ~ %
```

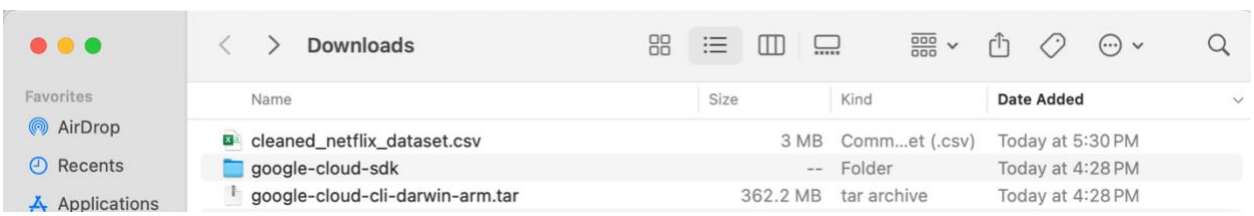
- **Download the File from GCP**:

We listed the files in the bucket to locate the dataset:

```
gsutil ls gs://netflix-eda-bucket/
```

Once located, we downloaded the file to my local system:

```
gsutil cp "gs://netflix-eda-bucket/cleaned_netflix_dataset.csv" ~/Downloads/
```



2. Uploading Data to Tableau Web

After successfully downloading the dataset from GCP, I manually uploaded it to Tableau Web for visualization.

- **Connect to Data in Tableau Web**

  Using the Tableau Web interface, we navigated to "**Connect to Data**" and selected **Text File** as the source. we then uploaded the cleaned_netflix_dataset.csv file from our local machine.

- **Verify and Prepare Data**

  Tableau displayed a preview of the data. Here, we confirmed that the column headers and data types were correctly identified.

Step 4: Tableau Dashboard Design

• With the dataset connected, we proceeded to create visualizations to analyze Netflix content trends effectively.

• Collaborative discussions shaped the story for the dashboard.

- **Build Visualizations**:

We used Tableau's drag-and-drop interface to create bar charts, pie charts, and other visualizations highlighting key metrics, such as content type distribution and ratings.

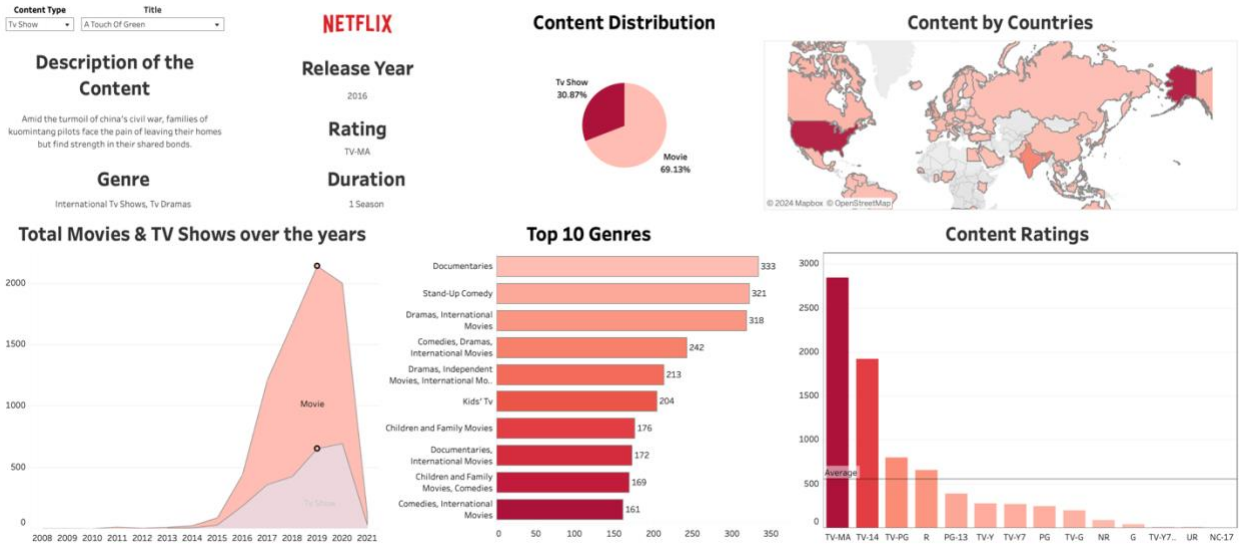
- **Design Dashboards**

After creating individual visualizations, we combined them into an interactive dashboard. This allows us to present insights clearly and effectively.

- **Save and Share**

Finally, we saved the dashboard on Tableau Web, making it accessible via a public link for easy sharing.

6. Tableau Dashboard Representation



Our team analyzed Netflix's content catalog using publicly available data. Thanks to free Tableau access provided through our Northeastern University accounts, we built an interactive dashboard to explore insights. The dashboard focuses on five key aspects to understand Netflix's content strategy better. Here's a step-by-step breakdown of the analysis process:

Content Distribution of Movies and TV Shows

Insight: Movies lead the catalog, accounting for 69.13% of the total content, compared to TV Shows at 30.87%. This significant difference highlights Netflix's continued focus on movies while steadily growing its TV show offerings to cater to binge-watching trends.

Content Distribution by Countries

Insight: Countries like the USA and India have the largest libraries, showing Netflix's global reach and regional focus.

Total TV Shows and Movies Added Over the Years
Insight: A sharp rise in original content since 2015 highlights Netflix's shift to content creation.

Top 10 Genres
Insight: Documentary, Stand-up comedies, and Drama, showing Netflix caters to varied audience preferences.

Content Ratings
Insights:

- The majority of Netflix content is rated **TV-MA**, catering to mature audiences, followed by **TV-14**, which appeals to teens and young adults.
- Family-friendly ratings like **TV-PG** and **PG** are also well-represented, though they trail behind more mature categories.
- Ratings such as **R** and **PG-13** reflect Netflix's effort to provide content for diverse age groups, balancing adult-oriented shows with offerings suitable for younger viewers.

7. References

[1] https://help.tableau.com/current/pro/desktop/en-us/examples_googlecloudstorage.htm
[2] https://help.tableau.com/current/pro/desktop/en-us/examples_textfile.htm
[3] https://public.tableau.com/en-us/s/
[4] https://pandas.pydata.org/docs/
[5] https://plotly.com/python/
[6]https://www.kaggle.com/datasets
[7] https://www.youtube.com/watch?v=BTArwS4ljC4
[8] https://www.youtube.com/watch?v=DyU51nRqcHg

8. Conclusion

This project highlights our team's collaborative efforts in analyzing Netflix's dataset to uncover meaningful insights into content trends and strategies. Through extensive exploratory data analysis (EDA) and effective visualization in Tableau, we identified patterns in genres, release periods, and audience preferences.

Overcoming challenges such as integrating Google Cloud Platform (GCP) with Tableau showcased our adaptability and technical expertise. This seamless integration of tools emphasizes our commitment to delivering comprehensive, data-driven solutions.

The project demonstrates how teamwork and technical skills can transform complex datasets into actionable insights, enabling informed decision-making through innovative visualizations.